	TOTAL FOR QUESTION 3	**7**
4(a)	Resistance of a circuit component is the ratio of the potential difference across the component to the current passing through it.	**B1**

Qns	Answer	Marks
4(b)	e.m.f. is the energy converted from non-electrical to electrical form per unit charge, while p.d. is the energy converted from electrical to non-electrical form per unit charge. [B1]	B1
4(c)	As R_2 increases, the total resistance of the circuit increases and the current in the cell decreases. The p.d. across the internal resistance therefore decreases. Hence, using $V_{terminal} = E - Ir$, the terminal p.d. increases. Since R_1 is constant, the current through R_1 increases. -	M1 M1 A1
4(d)(i)	p.d. across r = 1.40 − (0.220 × 5.50) = 0.19 V. Icell = 0.19/0.13 = 1.46 A ≈ 1.5 A (2 s.f.).	C1 A1
4(d)(ii)	I_2 = 1.5 − 0.220 = 1.28 A; R_2 = (0.220×5.50)/1.28 = 0.945 Ω	C1 A1
	TOTAL FOR QUESTION 4	9
5(a)	For a single-slit diffraction pattern,	